



Spatial Acoustic Analysis

Ryan Adolph, Michael Oswald, Micaylyn Parker

Introduction



Abstract

This project presents a modular, real-time audio sensing system using a Raspberry Pi to determine the direction and identity of acoustic signals. By integrating multichannel microphone input, beamforming, and FFT-based processing, the system generates compact frequency-domain profiles for analysis on an operator's PC. Designed for adaptability and low resource use, it supports dynamic environments and variable microphone arrays.

Background

Spatial awareness and signal profiling are increasingly important in both defense and civilian contexts, including surveillance, navigation, and environmental monitoring. Advances in low-cost hardware such as Raspberry Pi devices and USB-based audio interfaces have made it feasible to develop portable and scalable acoustic sensing systems. Techniques like beamforming and Fast Fourier Transform (FFT) analysis enable extraction of directional and frequency-domain information from complex audio environments.

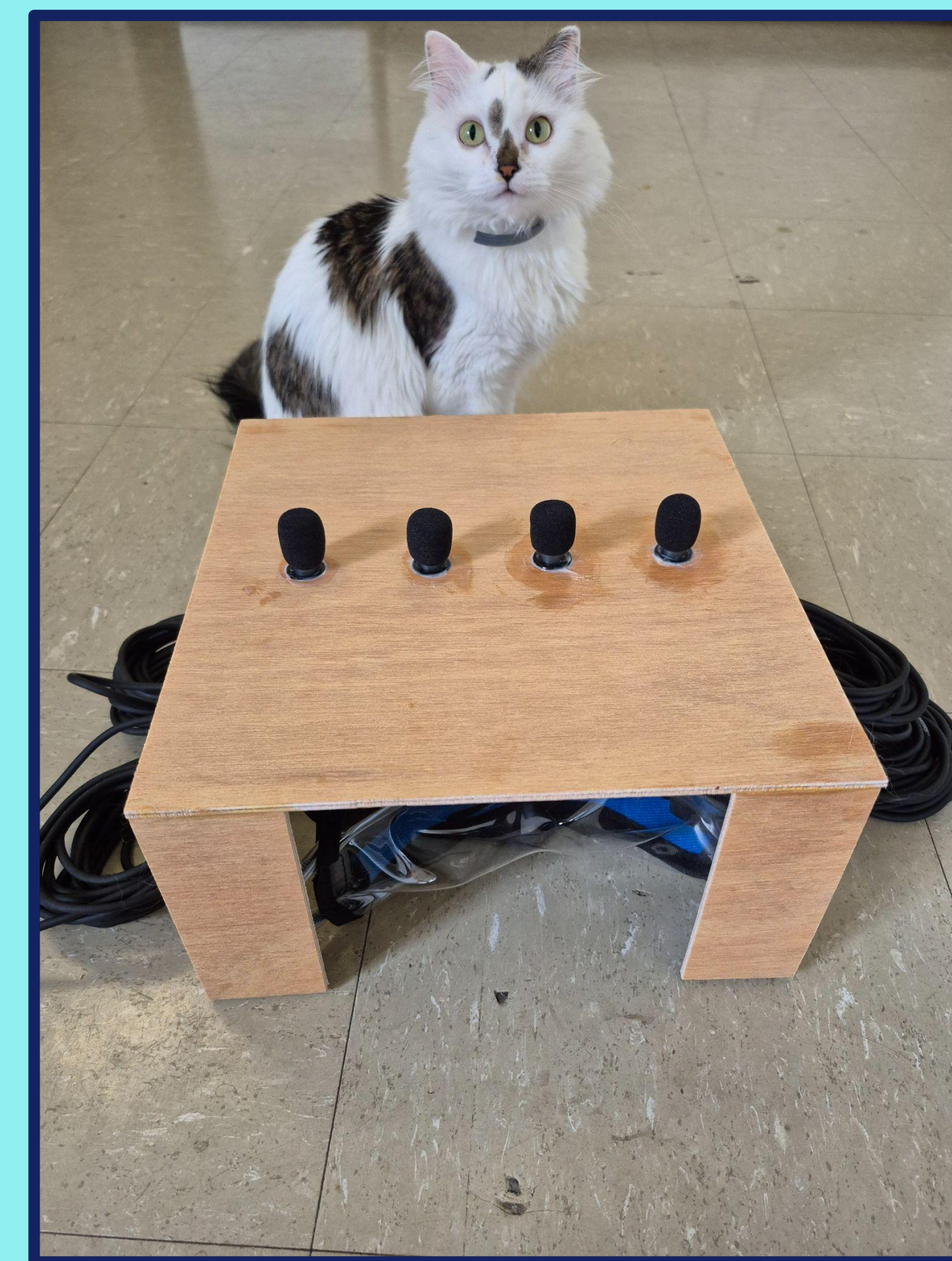
Objectives

- Design a modular audio sensing system supporting multiple microphone array configurations.
- Implement real-time audio acquisition and processing on a Raspberry Pi.
- Apply beamforming techniques for signal direction estimation.
- Utilize FFT to extract and compress frequency-domain features.
- Enable data transfer to an operator PC for signal comparison and profiling.
- Develop an interface for visualization and interpretation of signal profiles.

Development

Components

- Raspberry Pi 4 Model B
- Audio Interface (Mixpre 6 II)
- Microphones (4x Sure Beta 98)
- Connecting cables
- Operator PC

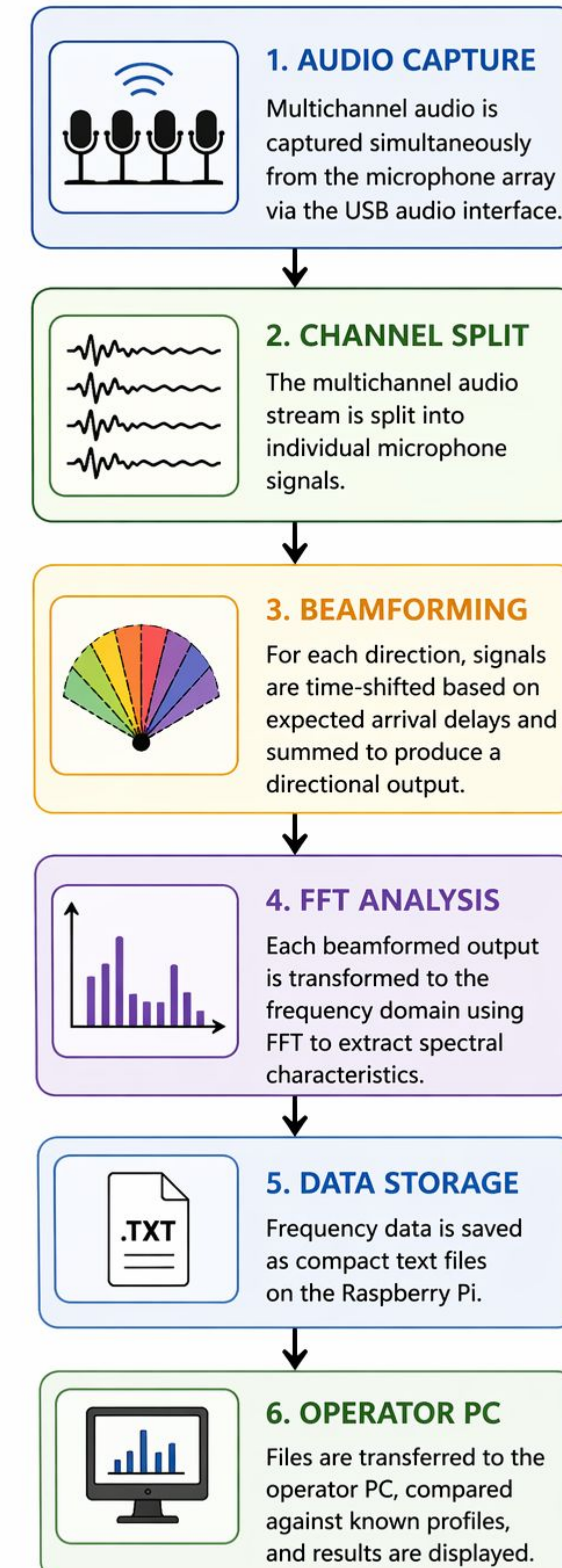


Methodology

A 4 microphone array captures audio in 0.25 second intervals every 0.5 seconds which outputs one large .wav file filled with 4 audio inputs. We then split the large .wav file into 4 separate .wav files to perform a beam forming analysis to locate which direction the audio signal was coming from. We then perform a FFT over the 4 .wav files to analyse each frequency. We then store the FFT output for the operators PC to come and grab to perform more signal analysis.

SIGNAL PATH

High-Level Overview



Conclusion

This project demonstrates the capability of small, affordable, and readily available components are capable of forming basic detection and alarm systems for applications such as drone defense. These components include the mathematical applications necessary to perform these analysis, making the need for mathematical or coding acumen minimal with the proper preparation. This follows the Ukrainian example of accessible drone defense within the reach of the public.

Opportunities for Expansion

This system provides the foundation for expansion into larger microphone systems; especially given a stronger computational center or FPGA processing of demanding processes such as beamforming and FFT analysis. Likewise, the use of an RF antenna capable of reading RF Frequencies can alert the system of when it should be observing new audio input. Multiple RF antennae has been a popular choice for similar detection arrays; but is a much more expensive and harder to access solution.

References

- Victor Anderssén, Computer Engineer and author of the FFT used: https://github.com/Avicted/c_fft_visualizer
- NumPy (Van der Walt et al., 2011)
- SciPy (Virtanen et al., 2020)
- Python 3 (Python Software Foundation)

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Contact Info

Ryan Adolph: ryanadolph7@gmail.com
Michael Oswald: oswald.contact@gmail.com
Micaylyn Parker: micaylynparker@gmail.com

Beamforming and Fourier Transforms

Fast Fourier Transforms are a streamlined (thus 'fast') method to break down our incoming audio into small, more manageable pieces. This reduces computational complexity from $O(n^2)$ to $O(n \log n)$.

Beamforming is a method that here phase shifts the incoming sound signals and compares them to selectively listen to a specific direction.

BEAMFORMING CONCEPT

Directional Listening Using Microphone Array

